Sign Problem and Complex Langevin Method

October 22, 2021

Summer Project Report

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Abstract

In this project, we explore the complex Langevin method, which was proposed for overcoming the notorious sign problem appearing in theories with complex actions. We look at the mathematical justification of this method and reasons of its failure in some cases. Later, we study the failure of the method in the parameter space of various one-dimensional models with the help of numerical simulations and check the validity of various correctness critarias for predicting the failure of simulations.

Acknowledgements

I would like to express my sincere gratitude to Dr. Anosh Joseph for giving me the opportunity to work under his guidance and for arranging regular discussions, which kept me motivated and made this summer project possible.

I would also like to thank Arpith Kumar for providing a code, which got me started with the project and Piyush Kumar for providing help during the course of project.

Contents

1 Introduction

An important object of interest in any field theory is the path integral weighted by the action of the theory, $\sim e^{-S(\phi)}$. The lattice formulation of the field theories helps us to non-pertubatively evaluate the path integrals and compute the required observables by the use of Monte Carlo approach as long as action is real. But we encounter many physical scenarios when action takes complex values like in heavy dense QCD, interacting Bose gas, and various other finite chemical potential systems. In such cases standard Monte Carlo approach, which assigns probability weight to the action part of path integral fails due to non-positive definiteness of the complex weight. This causes the *sign problem* due to incompatibility of Monte Carlo with complex probability weights. Complex Langevin method proposed by G. Parisi and J. Klauder [1] is a successful alternative in evading the sign problem, and has given correct results in most practical circumstances. However, being a numerical method, it also suffers with problems of runaways (overflows) and convergence to incorrect results [2, 3]. In this project, we looked into the complex Langevin method, studied the causes of its problems, and checked its reliability domain in some simple one-dimensional models.

In Section 2, we introduce briefly the complex Langevin method, stochastic quantization, and its origin from the real Langevin. After this, we will look at the equivalence of complex Langevin proposal to the original problem of path integral computation at hand and establish the mathematical justification for the same. Section 3 contains the results of various simulations done to study the failure of the method, and finally the conclusions and further outlook.

2 Complex Langevin Method

2.1 A Derivative of Real Langevin

Real Langevin, also called stochastic quantization is another way to quantize the field theory differing from canonical quantization. Here, field is taken as a stochastic variable, which evolves to the minimum of the action by the Langevin equation

$$\frac{d\phi(t)}{dt} = -\frac{\partial S[\phi]}{\partial \phi} + \eta(t). \tag{2.1}$$

Here, t is the Langevin time and $\eta(t)$ is the statistical noise obeying the condition

$$\langle \eta(t) \rangle_\eta = 0; \quad \langle \eta(t)\eta(t') \rangle_\eta = 2\delta(t - t'). \tag{2.2}$$

An equivalent way of studying Eq. (2.1) is its dual Fokker-Plank equation, which describes the evolution of probability density of the field given by

$$\partial_t P = L^T P; \qquad L^T = \nabla_\phi[\nabla_\phi + (\nabla_\phi S(\phi))]. \tag{2.3}$$

Now, when the process reaches equilibrium, we can write the expectation values of any observable as

$$\langle O \rangle = \int_{\mathbb{R}} O(\phi)P(\phi)d\phi. \tag{2.4}$$

For $(0 + 1)D$ case, the Langevin equation can be simulated using discretized form of Eq. (2.1)

$$dX = -S'(X)dt + dw; \qquad dw^2 = 2dt. \tag{2.5}$$

Here X is the field variable and dw is the added gaussian noise in the Langevin process.

Once the process has been thermalized, i.e., when the action has found its minima, one can evaluate the required observables as

$$\langle O(x) \rangle \simeq \frac{1}{N} \sum_{k=1}^{N} O(X_k). \tag{2.6}$$

In the cases when action is complex, the complex Langevin method proposes to complexify the field variables from $x \to z$ and $S(x) \to S(z)$. In which case, the complex Langevin equation for generating samples takes the form

$$dX = K_x dt + \sqrt{2N_R dt}, \tag{2.7}$$

$$dY = K_y dt + \sqrt{2N_I dt}, \tag{2.8}$$

with $K_x = -\mathrm{Re}(S'(z))$, $K_y = -\mathrm{Im}(S'(z))$. We have complexified the noise $\eta(t)$ to $N_R(t)+iN_I(t)$ such that $N_R - N_I = 1$. (This condition follows from Eq. (2.2)).

Expectation value of observable is now written as

$$\langle O(x+iy) \rangle \simeq \frac{1}{N} \sum_{k=1}^{N} O(X+iY). \tag{2.9}$$

Like for the real Langevin, we have a dual Fokker-Plank equation associated with complex Langevin process with the Fokker-Plank operator L_c^T given as

$$L_c^T = \nabla_x [N_R \nabla_x - K_x] + \nabla_y [N_I \nabla_y - K_y], \tag{2.10}$$

and

$$\langle O \rangle_c = \int_{\mathbb{R}} \int_{\mathbb{R}} O(x+iy) P(x,y) dx dy. \tag{2.11}$$

Thus, we see that the complex Langevin method evades the problem of complex probability weight by introducing higher dimensions and finding the equivalent positive probability weight in higher dimensional space.

2.2 Mathematical Justification

While it is clear that there is no physical justification of complexification of the field variables, we see that complex Langevin is also not fully justified in a mathematical sense as it requires the equivalence of equations (2.10) and (2.4), i.e.,

$$\begin{aligned} \langle O \rangle &= \lim_{t \to \infty} \left(\int_{\mathbb{R}} O(x) P(x;t) dx \right) \\ &= \lim_{t \to \infty} \left(\int_{\mathbb{R}} \int_{\mathbb{R}} O(x+iy) P(x,y;t) dx dy \right) \\ &= \langle O \rangle_c . \end{aligned} \tag{2.12}$$

It is to note that when action, S is complex, $P(x;t)$ will be a complex function. This is the reason one cannot use the standard Monte Carlo techniques as they lose meaning for complex probability weights. To prove the equivalence given in Eq. (2.22), Seiler has defined a function

$$F(t,\tau) = \int P(x,y;t-\tau) O(x+iy;\tau) dx dy. \tag{2.13}$$

We can see that $F(t,0) = \langle O \rangle_c$, and similarly it can be proved that $F(t,t) = \langle O \rangle$ as

$$
\begin{aligned}
F(t,t) &= \int P(x,y;0) O(x+iy,t) dx dy \\
&= \int P(x,y;0) \exp(tL) O(x+iy;0) dx dy \\
&= \int P(x;0) \exp(tL) O(x;0) dx \\
&= \int \left(\exp\left(tL^T\right) P(x;0)\right) O(x;0) dx \\
&= \int P(x;t) O(x;0) dx \\
&= \langle O \rangle.
\end{aligned} \tag{2.14}
$$

Here we make use of $P(x,y;0) = P(x;0)$ and assume that observables evolve by Kolmogorov equations [4, 5].

Our hope is to evaluate the derivative $\frac{\partial F(t,\tau)}{\partial \tau}$ and verify if it evaluates to 0, which will prove the equivalence of real and complex Langevin.

$$
\begin{aligned}
\frac{\partial F(t,\tau)}{\partial \tau} &= \frac{\partial}{\partial \tau} \int P(x,y;t-\tau) O(x+iy;\tau) dx dy \\
&= -\int \frac{\partial P(x,y:t-\tau)}{\partial \tau} O(x+iy;\tau) \\
&\quad + \int P(x,y;t-\tau) \frac{\partial O(x+iy;\tau)}{\partial \tau} dx dy \\
&= -\int L^T P(x,y;t-\tau) dx dy \\
&\quad + \int P(x,y;t-\tau) L O(x+iy,\tau) dx dy.
\end{aligned} \tag{2.15}
$$

One can observe that the two terms in the above integration vanish if we assume that the boundary terms do not contribute to the integral, which immediately justifies the complex Langevin process.

But it can be seen that it is not always the case. Assuming that the equilibrium probability distribution exists in the limit $\lim t \to \infty$, we have from Fokker-Plank equation, $L^T P(x,y;\infty) = 0$. Substituting it back into the equation and equating to zero, we obtain the equation for correctness for complex Langevin process,

$$
\frac{\partial F(t,\tau)}{\partial \tau} = 0 = \int P(x,y;\infty) L O(x+iy;\tau) dx dy. \tag{2.16}
$$

This shows that we require sufficient decay of the equilibrium probability distribution at large x, y for CLM results to be valid for a particular observable. This condition is oftentimes stated through the exponential decay of equilibrium probability distribution in imaginary direction.

It is to note that by above procedure we have not fully justified the complex Langevin process and the condition obtained in Eq. (2.13) is an observable specific condition, which may restrict the types of observables that can be computed correctly through complex Langevin procedure. For the detailed description taking into account the boundary terms one can look at Ref. [6].

3 Numerical Simulations

The simulations are done by starting from a random configuration point in the complex plane and evolving it using the discretized Langevin equations. Sufficient

Langevin time is given for field variable to thermalize to its equilibrium value. After thermalization, the values of field variable are collected and are used to calculate required observables.

Adaptive step sizes, dt are used to prevent runaways into large imaginary directions. The method used for calculating adaptive step size is taken from insert ref is briefly is follows:

1. Calculate the average of maximum drift values obtained over many thermalization, AVGDRIFT (or divide thermalization into different parts and take maximum values of drifts from each part),

2. Keep note of maximum drift value obtained during the evolution, MAXDRIFT and update it each Langevin step.

3. During each Langevin step, set step size $dt = \bar{d}t \times \frac{\text{AVGDRIFT}}{\text{MAXDRIFT}}$. Here $\bar{d}t$ is the cutoff on step size.

3.1 Criteria for Correctness

3.1.1 Localized Distribution of Probability

As concluded in section 3 that complex Langevin process is not fully justified and is correct only in the absence of boundary terms, therefore, for the reliability of results obtained through complex Langevin simulation, there are some correctness criteria, which are needed to be satisfied. The first one is localized distribution of equilibrium ϕ obtained in Eq. (2.16).

$$\int P(x, y, \infty) LO(x + iy, \tau) dxdy = 0 \tag{3.1}$$

This criteria is often checked by observing the distribution of drift terms, which are responsible for distribution of field variable, where drift terms refer to $v_{drift} = \left| -\frac{\partial S[\phi]}{\partial \phi} \right|$. If the decay of distribution is faster than exponential, complex Langevin results are considered reliable or any power decay is an indication of incorrect convergence of complex Langevin simulations.

3.1.2 Schwinger-Dyson Relations

The second criteria was introduced in [4] which is based on the Schwinger-Dyson relations for the n-point functions of the solutions. They read as,

$$C_O = \langle LO(\phi) \rangle = 0 \tag{3.2}$$

where $L = [\nabla_\phi - \nabla_\phi(S(\phi))] \nabla_\phi$. This equation simply tells the fact that expectation value of observables should not evolve once process has reached its equilibrium.

In general, these need to be satisfied for all observables $O(\phi)$. If we take observables as ϕ^n, the conditions becomes,

$$C_n = \frac{1}{n} \langle L\phi^n \rangle = 0 \tag{3.3}$$

for even n, as odd powers vanish trivially by symmetry. Since these need to be satisfied for all even n, just checking for n=2 does not guarantee the correct convergence. Usually, for numerical purposes checking upto n=6 is sufficient to ascertain the validity of simulations. These will be used to check the correctness of the complex langevin simulations in following sections.

3.2 Quartic Sigma Model

The action for quartic sigma model has the form

$$S = \frac{1}{2}(1 + iB)\phi^2 + \frac{1}{4}\phi^4, \tag{3.4}$$

where B is a real parameter and ϕ is taken as complex valued. Gradient of S is

$$\nabla S = (1 + iB)\phi + \phi^3. \tag{3.5}$$

After complexification of ϕ to $x + iy$ the drift terms take the form

$$K_x = -x + By - x(x^2 - 3y^2) \qquad K_y = -y - Bx - y(3x^2 - y^2). \tag{3.6}$$

The classical flow diagram corresponding to above drift values is shown in Fig. 1.

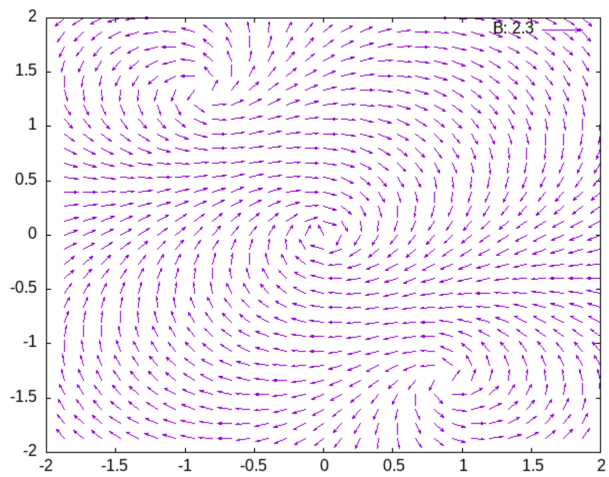
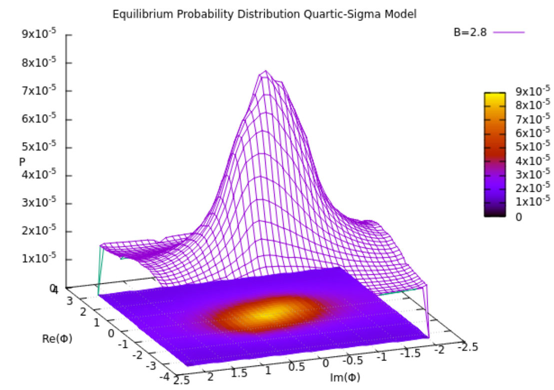


Figure 1: Negative gradient of the action is plotted to bring evident the classical trajectories in the background quartic-sigma action. We can also see the three critical points with stable at $\phi = 0$ and two unstable points corresponding to $\phi = \sqrt{1 + iB}$.

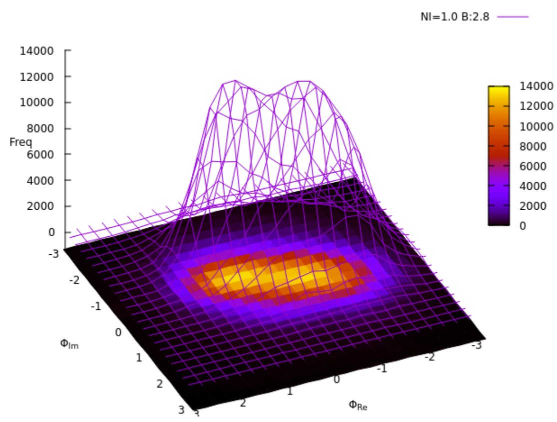
The discretized evolution equations for complex Langevin with real noise are

$$dX = K_x dt + \sqrt{2N_R dt} \tag{3.7}$$
$$dY = K_y dt \tag{3.8}$$
$$\langle N_R(t)N_R(t')\rangle = 2\delta(t, t'). \tag{3.9}$$

Equilibrium distribution of ϕ obtained after thermalization is shown in Fig. 2. It shows that the comparison between distributions at $N_I = 0$ and $N_I = 1$.



(a) N_I (complex noise) $= 0$.



(b) N_I (complex noise) $= 1.0$.

Figure 2: Equilibrium disribution of field variable ϕ.

The results obtained by simulation of quartic sigma model are presented in figures shown below. Fig. 3 shows the probability distribution of the drift terms in the imaginary direction for different values of the parameter B. Here, we record the shift from faster than exponential decay to power law decay as B crosses the value 2.8, indicating the incorrect convergence of complex Langevin. This is verified in Fig. 4, where the observable ϕ^2 is plotted against the various values of B and we can observe the indications of incorrect convergence of CLM around $B = 2.8$ as predicted by the drift decay distribution in Fig. 3.

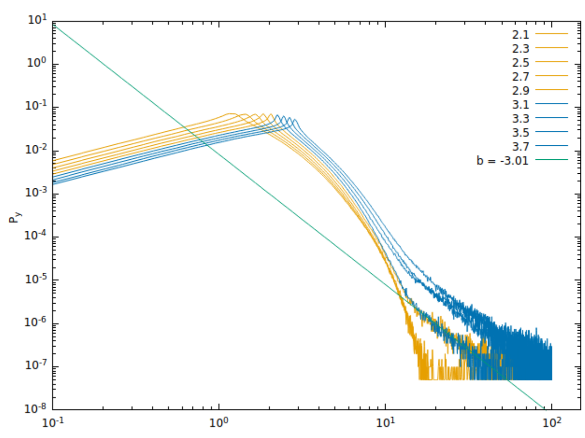
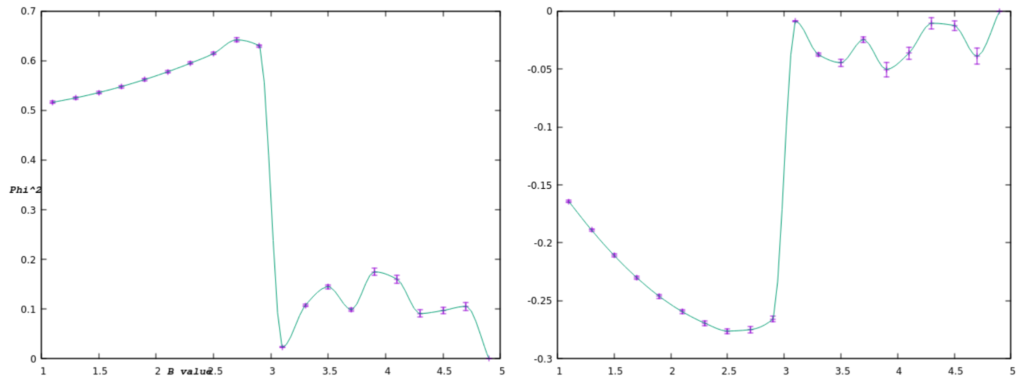
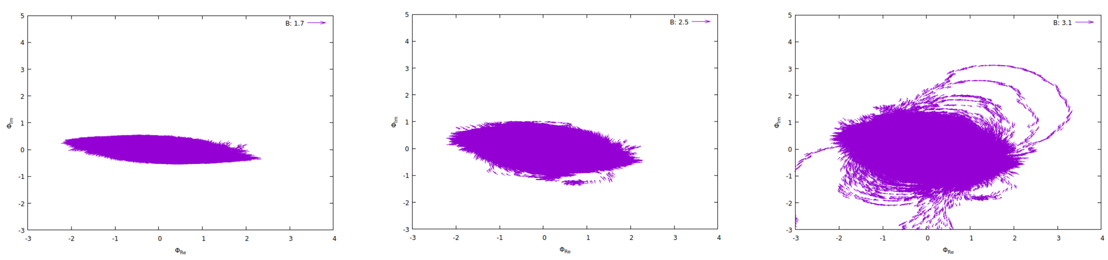


Figure 3: Decay of drifts for different values of parameter B. We can see power decay starting from B value of 2.7 indicating the breakdown of complex Langevin process.



(a) ϕ^2 vs. B. (b) $\mathrm{Im}(\phi^2)$ vs. B.

Figure 4: Observable ϕ^2 and its imaginary part against different values of B for quartic sigma model. Incorrect convergence can be seen near the B value of 2.8.



(a) $B = 1.7$. (b) $B = 2.5$. (c) $B = 3.1$.

Figure 5: Equilibrium distributions of at different values of B.

Fig. 5 shows the equilibrium distributions at different values of B. It can be observed that below $B = 2.5$, the distribution is localized, bounded in the imaginary direction. But as B value increases beyond 2.5, the bound in the imaginary direction vanishes resulting in a delocalized distribution. This delocalization is suspected as the cause of incorrect convergence of complex Langevin simulations. This delocalisation is happening because of the presence of two unstable critical points in Fig. 1. The bounds in the imaginary directions are analytically calculated by Aarts in Ref. [4], where it was shown that bound vanishes in the quartic sigma model for $B > 1.7$.

3.3 The Guralnik Pehlevan Model

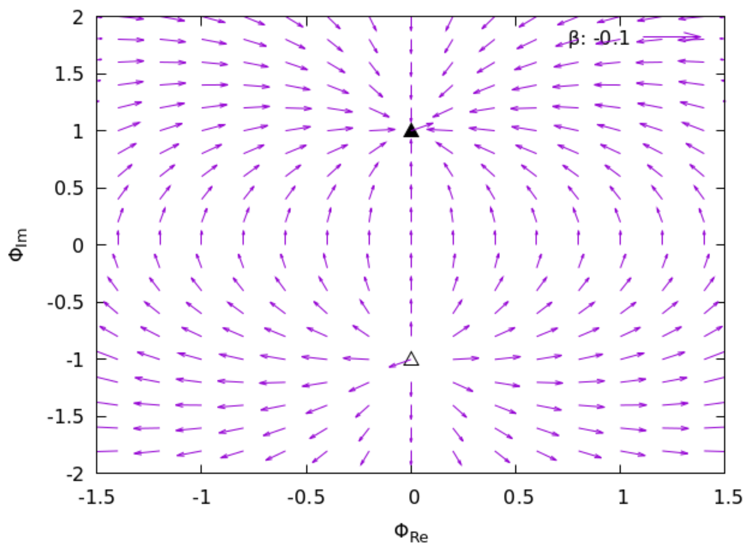
The action for the model is

$$S = i\beta \left(\phi + \frac{\phi^3}{3} \right). \tag{3.10}$$

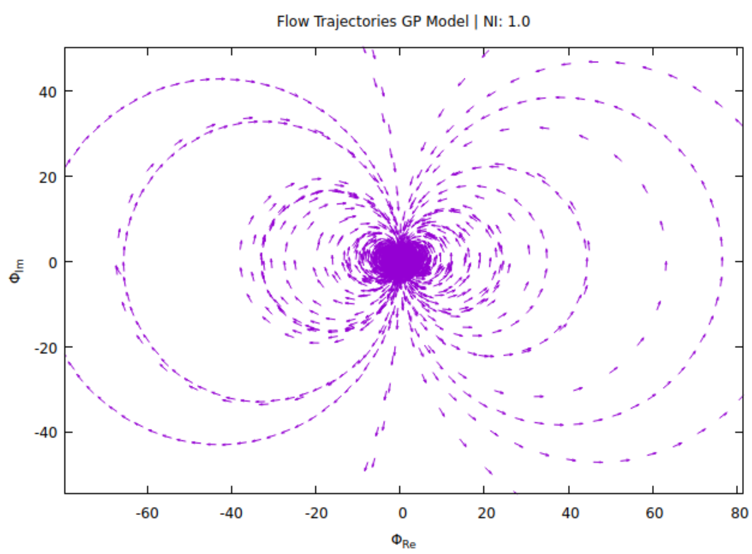
The gradient of the action is written as

$$K = i\beta \left(1 + \phi^2 \right). \tag{3.11}$$

The critical points $\pm i$ and gradient flow is shown in Fig. 6a. Fig. 6b shows the classical trajectories of ϕ obtained by complex Langevin simulation at complex noise, $N_I = 1$.



(a) Two critical points of GP action at $z = \pm i$.



(b) Classical flow trajectories obtained from simulation.

Figure 6: Critical points and classical flow trajectories of the Guralnik Pehlevan model.

Next we check the correctness of complex Langevin results for increasing value of complex noise by observing decay of the drift distributions. The drift decays obtained for different complex noises are shown in Fig. 7. The power law decay becomes apparent after $N_I = 0.3$ predicting incorrect convergence of complex Langevin simulations. The predictions from drift decay is verified in Fig. 8 where also deviations from correct results is seen when $N_I > 0.3$.

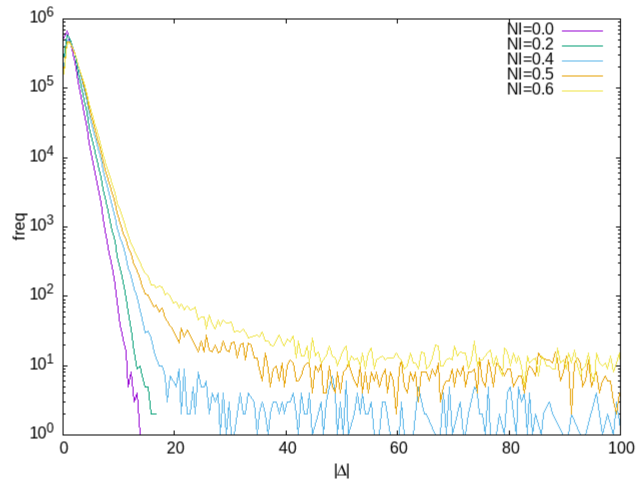


Figure 7: The power law decay of drift can be observed beyond $N_I = 0.3$ indicating incorrect convergence of complex Langevin simulation.

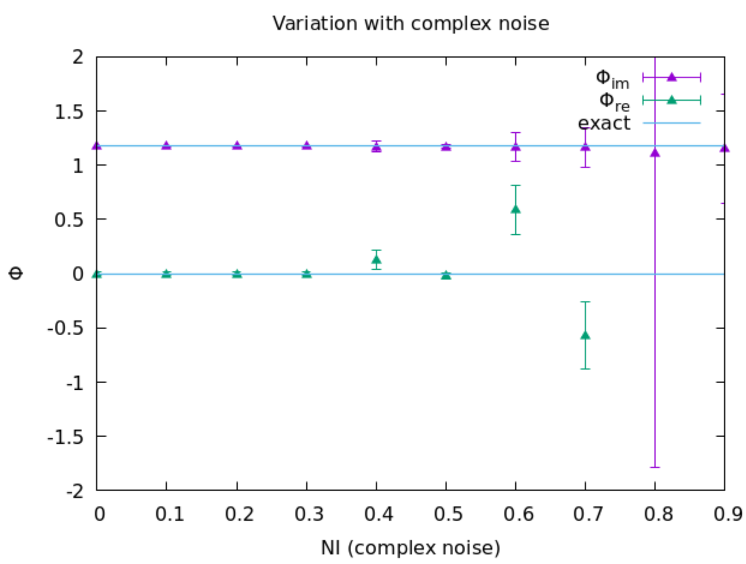
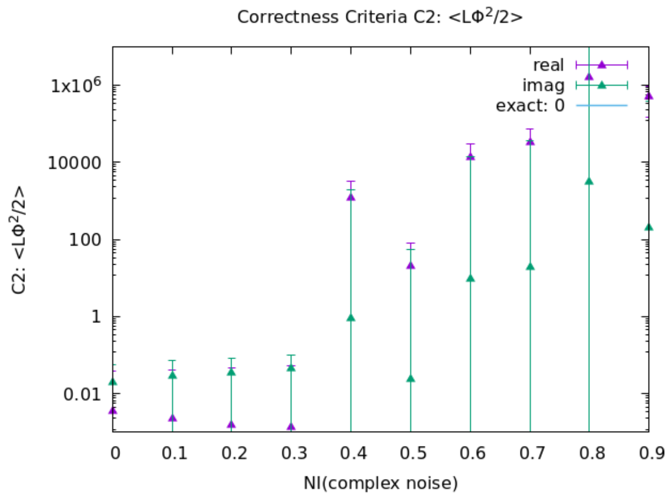
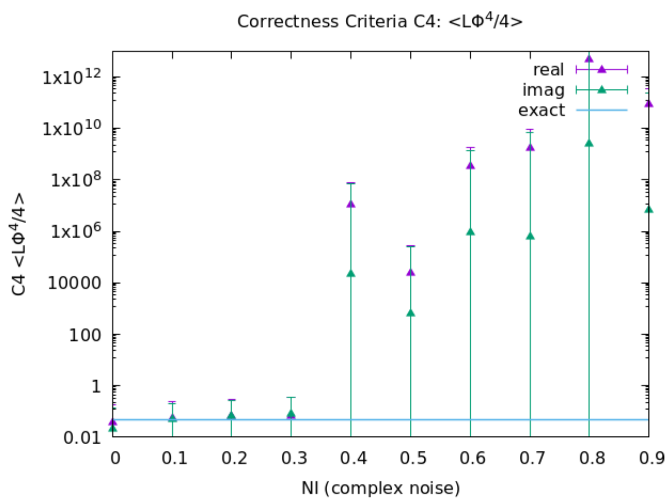


Figure 8: The exact and calculated results using complex Langevin. The correctness criteria predictions are verified as complex Langevin values of ϕ start to deviate from exact results beyond $N_I = 0.3$.



(a) C_2 moment for observable ϕ.



(b) C_4 moment for observable ϕ.

Figure 9: C_2 and C_4 moments for observable ϕ showing the breakdown of complex Langevin simulation beyond $N_I = 0.3$.

Next, we use the second correctness criteria intoduced in section 3.1.2 to check its validity in predicting failure of simulations. Results are shown in Fig. 9 , showing the deviations in C_2 and C_4 moments starting at $NI > 0.3$ indicating use of Schwinger-Dyson relations as valid criteria for correctness.

4 Conclusions

We analyzed two zero-dimensional models with sign problem using complex Langevin method. The correctness of simulations was checked using two criteria for correctness, the decay rate of the drift distribution and Schwinger-Dyson relations. It was observed that both of the correctness criteria are able to correctly predict the failure of complex Langevin, But moments corresponding to Schwinger-Dyson relations are observed to show large error bars, which limit the predictability of failure and is thus seemingly less reliable based on the other models which were analyzed.

Major Cause for incorrect convergence of Complex Langevin Simulations is found to be the large runaways of the samples due to higher values of imaginary component of drift terms which is responsible for the non-vanishing Boundary terms. The problem of large runaways prevents system from thermalising and thus leading to incorrect results. In the specific models analysed, the problem of runaways was found to be associated with the unstable zeros of drift terms.

5 Further Goals

After being acquainted with the machinery of complex Langevin and understanding the theory related to the boundary terms leading to failure of the complex Langevin methods, the immediate goals are the following: calculate and analyze the boundary terms using the complex Langevin simulations, study the regularization of complex Langevin [7]. It would also be interesting to use the the method of complex Langevin in problems related to equation of state of dense matter [8, 9].

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